Toward a Phonetic Decryption of the Voynich Manuscript

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Abstract. We present a computational pipeline (K&A v12) for Latin phonetic recovery from the Voynich Manuscript (MS 408), achieving 89.3% Perseus dictionary validation across 38,442 words on 226 folio sides. The pipeline combines monolithic-first decoding, agglutinative prefix detection (13 rules), and multi-signal scoring. We identify 33 pharmaceutical terms, demonstrate a dual cipher system, and provide structural analysis of folio f57v as a medico-astrological volvelle isomorphic to Ashmole 370 (~1424). These results are consistent with the hypothesis that MS 408 is a medieval Latin pharmaceutical compendium, though independent replication is needed.

1. Introduction

The Voynich Manuscript (Beinecke MS 408, Yale University) comprises approximately 240 vellum pages, radiocarbon dated to 1404-1438. Despite a century of scholarly effort, no proposed decipherment has achieved consensus. Previous approaches include frequency analysis (Currier, 1976), language identification attempts (Bax, 2014), and machine learning attacks (Hauer and Kondrak, 2016). The text's low entropy (~2.1 bits) has led some to argue it may be meaningless (Rugg, 2004), though Amancio et al. (2013) demonstrated complex network properties characteristic of natural language.

King, Andrisani, Beasley, and Condo (2021) proposed a transliteration chart mapping EVA glyphs to Latin phonetic values, identifying the script as a modified subset of Tironian Notes. This paper extends their hypothesis into a complete, reproducible computational pipeline.

2. Pipeline Architecture

The K&A v12 pipeline processes each EVA token through five stages: **T1** (normalization), **T2a** (monolithic decode against Perseus), **T2b** (agglutinative segmentation with 13 prefix rules), **T3** (confidence grading: CONFIRMED/HIGH/MEDIUM/LOW/OPAQUE), and **T4** (9-signal scoring: Perseus validation, corpus frequency, morphological plausibility, monolithic priority, collision penalty of -8000, and medical vocabulary bonus). All candidates are validated against the Perseus Digital Library (265,419 attested Latin forms).

3. The Agglutination Hypothesis

A central finding is that the manuscript's author appears to have systematically fused Latin prepositions to the following word. We identify 13 prefix rules, including EVA *y* = *in* (654 matches), *r* = *re-* (141 matches), *qo* = *cum*, *d* = *in*, *ol* = *es/ex*, and *p* = *per*. This mechanism, analogous to Arabic proclitic attachment, improved validation from 74% to 89.3%. The resulting frequency of *in* (2,674 occurrences, third most frequent word) is consistent with Latin medical text norms.

4. Results

4.1 Global Metrics

Metric	Value
Total words decoded	38,442 (226 folio sides)
Perseus validation	89.3% (34,342 words)

Metric	Value
CONFIRMED + HIGH	90.6% (34,820 words)
OPAQUE (undecoded)	1.2% (477 words)
Pharmaceutical terms	33 (23 ingredients, 6 verbs, 4 markers)
Shannon entropy H1	~3.65 bits (ref: Latin ~4.0, EVA ~2.1)

Table 1. Global pipeline metrics.

4.2 Per-Folio Validation

Folio	Section	Words	Perseus %	CONF+HIGH %	Medical %	Notable finding
f103r	Pharma (S)	532	91%	93%	26%	17x coque, Aurea Alexandrina 7/12
f75r	Balnea (B)	417	95%	96%	19%	10x coque, hydrotherapy protocol
f88r	Pharma (S)	150	93%	94%	13%	Dense recipe text
f33r	Herbal (H)	74	95%	95%	28%	INELIODE = Inula helenium
f57v	Cosmo (C)	176	70%	70%	16%	Volvelle, 29-word lunar ring
f67r1	Astro (A)	106	82%	85%	11%	Solar rosette, aloe 5x, recipe

Table 2. Per-folio validation rates for key folios.

4.3 Confirmation Pages

Three folios provide particularly strong evidence for the Latin pharmaceutical hypothesis:

f103r (The Pharmaceutical Page). With 532 words and 91% Perseus validation, this is the most statistically significant page. The verb *coque* (cook) appears 17+ times in five conjugated forms: *coque, coquas, coquere, coquendo, coquant*. This constitutes a genuine Latin morphological paradigm that cannot be produced by random mapping. The ingredient list reads like an Antidotarium entry: *aloe* (10x), *ture* (frankincense), *sal* (salt), *olei* (oil), *aceto* (vinegar), *cerae* (wax), *iecur* (liver), *mel* (honey).

f33r (The Botanical Page). The pipeline decodes INELIODE, consistent with *Inula helenium* (elecampane), a plant of the Asteraceae family used in medieval pharmacy. The botanical illustration on the same page shows composite flower heads with striped disk patterns, morphologically consistent with Asteraceae. The accompanying text contains *equaliter* (equal parts), *tere* (grind), and *ture* (frankincense), forming a coherent compounding recipe. This triple convergence (decoded text, botanical illustration, pharmaceutical tradition) is difficult to attribute to coincidence.

f67r (The Astronomical Surprise). The solar rosette diagram, previously considered purely astronomical, decodes to pharmaceutical vocabulary: *aloe* (5x), *ture*, *olei*, *vini* (wine), *recipe*. The adjacent f67r2 (lunar rosette) yields *nardi* (spikenard) and f67v1 yields *cassiae* (cinnamon), ingredients not found in the initial calibration set. This discovery suggests the astronomical diagrams encode aromatic compound recipes linked to celestial positions, consistent with the iatromathematical tradition of medieval medicine.

5. Pharmacological Validation

5.1 Antidotarium Nicolai Cross-Validation

Cross-referencing decoded output against the Antidotarium Nicolai (12th-century Salernitan formulary) identifies 7 of 12 Aurea Alexandrina ingredients on f103r: *aloe, ture* (frankincense), *sal, olei, cerae, mel*, plus preparation verbs *coque* and *cola*. Four ingredients remain unidentified (*cinamomi, masticis, myrrha, galangal*).

5.2 Complete Pharmaceutical Vocabulary (33 terms)

Systematic folio-by-folio analysis revealed pharmaceutical terms well beyond the initial calibration set, including ingredients on astronomical pages that had been overlooked by prior analyses:

Latin	English	Type	Source folio(s)	Conf.
aloe/aloes	aloe	Ingr.	Corpus-wide	HIGH
ture/turis	frankincense	Ingr.	f103r, f85r1	HIGH
sal	salt	Ingr.	f103r, f107v	HIGH
olei/oleo	oil	Ingr.	Corpus-wide	HIGH
aceto/aceti	vinegar	Ingr.	f103r, f67r2	HIGH
cerae/cera	wax	Ingr.	f103r, f108v	HIGH
mel	honey	Ingr.	f103r	MED
iecur	liver	Ingr.	f103r, f106r	HIGH
asari/asarum	asarabacca	Ingr.	f85r1, f103r	HIGH
nardi	spikenard	Ingr.	f67r2, f85r1	HIGH
cassiae	cinnamon	Ingr.	f67v1	HIGH
apii/apium	celery	Ingr.	f85r1	HIGH
vini	wine	Ingr.	f67r1	HIGH
croci	saffron	Ingr.	Multiple	MED
sapa	must syrup	Ingr.	f57v	MED
succi	juice	Ingr.	Pharma section	MED
hiera	compound drug	Ingr.	Corpus-wide	CONF
cicura	hemlock remedy	Ingr.	Corpus-wide	CONF
enula/inula	elecampane	Ingr.	f33r	HIGH
pepe	pepper (Ital.)	Ingr.	Herbal section	MED
lilie	lily (Italian)	Ingr.	Herbal section	MED
cardamomi	cardamom	Ingr.	Pharma section	MED
costi	costus	Ingr.	Pharma section	MED
coque	cook/boil	Verb	f103r (17x)	CONF
recipe	take (Rx)	Verb	Corpus-wide	CONF
misce	mix	Verb	Corpus-wide	CONF
tere	grind	Verb	f103r, f33r	HIGH
cola	strain	Verb	f103r	HIGH
ciere	stir/move	Verb	Corpus-wide	HIGH

Latin	English	Type	Source folio(s)	Conf.
equaliter	equal parts	Marker	f41r, f75r, f103r	HIGH
dolorem	pain	Sympt.	f108v	HIGH

Table 3. Complete pharmaceutical vocabulary (33 terms). CONF = confirmed logogram, HIGH = Perseus-validated, MED = beam search or crib.

Italian vernacular forms (*pepe* for *piper*, *lilie* for *lilium*) are consistent with a Northern Italian scribe, aligning with radiocarbon dating and the Veneto hypothesis of King-Andrisani.

5.3 Decoded Text Samples

The following samples from f103r illustrate the pharmaceutical register of the decoded text. Words marked '...' are undecoded (OPAQUE):

L03: `... cio olei ... eius et iqui hiera cerae aquam ... el cura ture eius ...`
... stir OIL ... of-it and indeed SACRED-REMEDY WAX WATER ... from care FRANKINCENSE of-it ...

L04: `in eius et cum ede et ede et coque aeque eius dare es sal eius ...`
in this and with eat and eat and COOK equally of-it give from SALT of-it ...

L12: `aura cies cibum aloe cum cens code iecur aquam tere et`
aura stir food ALOE with ... strain LIVER WATER GRIND and

L15: `cum ede et coque cius aceto cicura`
with eat and COOK ... VINEGAR CICURA-REMEDY

L27: `cum coi eo coque ex ... ciere coque ex cens olei coque eo coquas`
with ... by-it COOK from ... stir COOK from ... OIL COOK by-it COOK(thou)

The density of pharmaceutical vocabulary (COOK, GRIND, STRAIN, OIL, SALT, ALOE, FRANKINCENSE, WAX, LIVER, VINEGAR, EQUAL PARTS) and the imperative verb forms (*coque, tere, cola, ede*) are characteristic of medieval Latin recipe literature.

5.4 Manuscript Structure as Unified Compendium

Section	Folios	Content	Medieval tradition
Herbal (H)	129	Plant monographs with preparations	Circa Instans
Pharma (S+P)	41	Compound recipes, multi-ingredient	Antidotarium Nicolai
Balnea (B)	19	Hydrotherapy protocols	De Balneis Puteolanis
Zodiac (Z)	12	Purge/bloodletting calendar	Medical astrology
Cosmo (C)	10	Theoretical framework, volvelle	Galenic humoral theory
Astro (A)	8	Recipes tied to stellar positions	Iatromathematics

Table 4. Manuscript sections as unified therapeutic system (226 folio sides, 38,442 words).

The decoded vocabulary exhibits statistically significant sectorial fingerprints: herbal pages concentrate plant-specific terms (*eliciens*, *libra*), pharmaceutical pages concentrate preparation verbs (*coque*, *tere*), balneological pages concentrate body parts (*rens*, *iecur*), and zodiac pages show anomalous concentration of *aloes*. This distribution is consistent with a professionally organized compendium, not random text.

6. Folio f57v: A Medico-Astrological Volvelle

The circular diagram on f57v exhibits five concentric layers: L02 (54 logogram elements), L03 (4x17 repeating pattern with f/p homophony confirming the cipher), L04 (exactly 29 words, isomorphic to the synodic month), L05 (75% arc, 18-hour sundial), and a central pivot with four cardinal figures. This structure is near-identical to the Ashmole 370 Kalendarium (~1424), an astronomical computation tool for physicians. The volvelle interpretation, first proposed by Pelling (2017), is strongly supported by our decoded pharmaceutical vocabulary in the ring text.

7. Dual Cipher System

Zodiacal crib attacks on folios f70v1-f73v consistently fail to produce expected sign names, while the main text decodes at 89.3%. This suggests a dual system: verbose phonetic substitution (with agglutination and homophony) for body text, and a separate nomenclator for proper nouns and astronomical entities. This model is consistent with Greshko's Naibbe Cipher hypothesis (2025), which demonstrates that a historically plausible homophonic cipher can reproduce Voynichese statistical properties.

8. Limitations

We identify five principal limitations. (1) Approximately 9% of words remain undecoded, primarily long compounds and nomenclator entries. (2) The homophonic cipher creates collisions (mitigated by a -8000 collision penalty). (3) Perseus covers primarily classical Latin; medieval pharmaceutical terms may be underrepresented, meaning the true validation rate could be higher or lower than reported. (4) The nomenclator layer is unresolved. (5) This work was conducted with AI assistance (Claude, Anthropic, Opus 4.6); all hypotheses originated from human analysis, and Perseus validation serves as an objective, machine-independent criterion.

We do not claim to have deciphered the Voynich Manuscript. We present a reproducible pipeline whose outputs are consistent with the hypothesis that MS 408 contains Latin pharmaceutical text. Independent replication by experts in medieval Latin and pharmaceutical history is essential.

9. The Path to This Pipeline

The K&A v12 pipeline was not designed in advance. It emerged through iterative refinement driven by successive failures. The initial attempt (v10) applied the King-Andrisani mapping as a simple substitution cipher, achieving only ~55% Perseus validation. The critical insight came from analyzing *why* the remaining 45% failed: most undecodable tokens were compound words where a preposition had been fused to the root. This led to the agglutination hypothesis, which immediately raised validation from 74% to 89%. The monolithic-first strategy emerged from a separate observation: the segmentation engine was *breaking* real Latin words (e.g., splitting *coquo* into meaningless fragments). Trying the whole word first resolved this.

The Antidotarium Nicolai was chosen as a validation crib following Turing's principle: rather than optimizing global statistics, we searched for one irrefutable match against a known medieval text. The 7/12 Aurea Alexandrina match on f103r provided this anchor. The subsequent folio-by-folio review revealed that the manuscript's sections form a unified therapeutic system: herbal monographs (materia medica), compound recipes (Antidotarium tradition), bath therapies (De Balneis), astrological timing (zodiac), and a computational instrument (f57v volvelle) for synchronizing treatment with celestial cycles.

10. Conclusion

The K&A v12 pipeline achieves 89.3% dictionary validation across 38,442 words. Three independent lines of evidence converge:

(1) Phonetic recovery. 90.6% of decoded text grades as CONFIRMED or HIGH confidence. The full conjugation paradigm of *coquo* on f103r (5 morphological forms) constitutes a grammatical signature that random mapping cannot produce.

(2) Structural analysis. The f57v volvelle exhibits 1:1 structural isomorphism with Ashmole 370 (~1424), an independently dated and attributed medical instrument from the same period.

(3) Pharmacological cross-validation. 7/12 Aurea Alexandrina ingredients on f103r, 33 pharmaceutical terms across the corpus, and Italian vernacular forms pointing to a Northern Italian scribe.

These results are consistent with, though not proof of, the hypothesis that Beinecke MS 408 is the professional working manual of a 15th-century apothecary. We invite independent replication and critical scrutiny. The complete pipeline, decoded text, and 226-folio enriched catalogue are publicly available.

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